

PROPHECY TRADING PLATFORM

Operations, Execution, Testing & Production Deployment Guide

Overview: Prophecy is an enterprise-grade options trading, backtesting, and paper-execution platform engineered for Indian derivatives (NIFTY / BANKNIFTY). It features multi-timeframe strategy evaluation across 5 synchronized timeframes (1m, 3m, 5m, 15m, 30m), an authoritative technical indicators engine, automated option contract selection, pre-trade risk management with emergency kill-switch protection, and a real-time FastAPI + Server-Sent Events (SSE) web dashboard.

1. Prerequisites & System Setup

- **Python Version:** Python 3.9+ (Python 3.10/3.11/3.12 fully supported).
- **Package Management:** uv (recommended) or standard pip/venv.
- **Operating System:** Windows 10/11, Linux (Ubuntu 20.04+), or macOS.
- **Repository Clone:**

```
git clone https://github.com/skynetukhra-coder/dev_stock_01.git
cd dev_stock_01
```

2. Running Locally

Method A: Using uv (Fastest, zero manual venv configuration)

PowerShell (Windows):

```
$env:PYTHONPATH = "engine/src;backend"
uv run --with fastapi --with uvicorn --with pydantic uvicorn app.main:app --host 127.0.0.1 --port 8000
--reload
```

Bash (Linux / macOS):

```
export PYTHONPATH="engine/src:backend"
uv run --with fastapi --with uvicorn --with pydantic uvicorn app.main:app --host 127.0.0.1 --port 8000
--reload
```

Method B: Using Standard Python Virtual Environment

```
python -m venv .venv
source .venv/bin/activate # On Windows: .venv\Scripts\Activate.ps1
pip install fastapi uvicorn pydantic httpx
export PYTHONPATH="engine/src:backend"
uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

3. Web Dashboard & Testing Strategy Cases 1 to 6

Once launched, navigate to <http://127.0.0.1:8000/dashboard> to access the real-time operator matrix.

Case #	Strategy Name	5-Timeframe Condition Rule	PCR Rule	Resolved Contract
Case 1	Straddle Baseline	ADX 15-30 & RSI 40-60 on 1m, 3m, 5m, 15m, 30m	PCR <= 0.60	ATM CE + ATM PE

Case 2	Straddle Special	ADX 0-10 & RSI 40-60 on all 5 timeframes	PCR <= 0.50	ATM CE + ATM PE
Case 3	Directional Call	ADX 15-30 & RSI 0-60 on all 5 timeframes	PCR < 0.80	ATM/ITM-1 CE
Case 4	Directional Put	ADX 15-30 & RSI 45-100 on all 5 timeframes	PCR > 1.25	ATM/ITM-1 PE
Case 5	Call Special	RSI <= 25.0 (extreme oversold) on all 5 timeframes	Extreme Low	ATM/ITM-1 CE
Case 6	Put Special	RSI >= 70.0 (extreme overbought) on all 5 timeframes	Extreme High	ATM/ITM-1 PE

4. Automated Verification & Quality Assurance

The platform carries **100 automated unit and integration tests** with a 100% pass rate:

```
# 1. Run 93 Engine Unit Tests (indicators, strategy cases, contract selector, risk manager)
uv run python -m unittest discover -s engine/tests -v

# 2. Run 7 FastAPI Integration Tests (REST endpoints, SSE events, kill switch, backtester)
uv run --with fastapi --with uvicorn --with pydantic --with httpx python -m unittest discover -s backend/app/tests -v

# 3. Run PEP-8 & Linter Quality Checks
uv run --with ruff ruff check engine/src engine/tests backend/app
uv run --with ruff ruff format --check engine/src engine/tests backend/app
```

5. Environment & Safety Parameters (.env)

```
PROPHECY_EXECUTION_MODE=MANUAL_CONFIRMATION # SIGNAL_ONLY | MANUAL_CONFIRMATION | AUTO_PAPER | LIVE
PROPHECY_MAX_DAILY_REALIZED_LOSS=10000.0 # Max daily loss ceiling in INR
PROPHECY_MAX_DAILY_TOTAL_LOSS=15000.0 # Max total drawdown ceiling in INR
PROPHECY_MAX_OPEN_POSITIONS=3 # Max concurrent open positions
PROPHECY_MAX_TRADE_NOTIONAL=50000.0 # Max rupee notional per trade
PROPHECY_LIVE_TRADING_ACKNOWLEDGED=FALSE # Mandatory live gate flag
PROPHECY_OPERATOR_SIGNATURE= # Live operator cryptographic authorization
```

6. Production Deployment Architectures

Option 1: Containerized Docker Deployment (Recommended)

```
# Build Docker image
docker build -t prophecy-engine:latest -f infra/docker/Dockerfile .

# Run container with auto-restart and environment mapping
docker run -d --name prophecy-app -p 8000:8000 --env-file .env --restart unless-stopped prophecy-engine:latest
```

Option 2: Linux Systemd Daemon + NGINX (with SSE Buffer Bypass)

NGINX reverse-proxy configuration (/etc/nginx/sites-available/prophecy):

```
server {
    listen 80;
    server_name trading.yourdomain.com;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

```
# CRITICAL: Disable buffering for Server-Sent Events (SSE) real-time streaming
proxy_buffering off;
proxy_cache off;
proxy_set_header Connection '';
proxy_http_version 1.1;
proxy_read_timeout 86400s;
}
```

7. Operational Runbooks & Emergency Procedures

- **Emergency Kill Switch:** In case of abnormal exchange volatility, click **STOP KILL SWITCH** on the web dashboard or send `POST /kill-switch {"active": true}`. All new entries are blocked instantly while liquidation exits remain active.
- **Loss Limit Lockdown:** If daily realized loss exceeds █10,000, the Risk Manager automatically transitions to lockdown until the next trading session (09:15 IST).
- **6-Lock Live Execution Gate:** Real broker orders require multi-key authorization, single-use token nonces (60s TTL), size ceilings (max 100 units / █35,000), and 10% fat-finger price sanity checks.